

Multi Agent Workflow Design Doc

The system is a multi-agent workflow built in Google Opal using a fixed pipeline that is as follows:

Inputs → Planner Agent → Monitoring Agent → Replanning Agent → Critic Agent → Output

Planner Agent

Role: Creates initial academic plan

Input:

- Course name
- Assignment name
- Due date
- Assignment instructions
- Rubric
- Estimated study time

Output:

- Task breakdown
- Timeline
- Milestones
- Estimated effort
- Risks

Monitoring Agent

Role: Evaluates the feasibility of the plan

Input:

- Planner agent output

Output:

- Status
- Scheduling issues
- Workload issues
- Recommendations

Replanning Agent

Role: Revises the plan if issues are found

Input:

- Planner agent output
- Monitoring agent output

Output:

- Revised plan
- Updated timeline
- Changes made
- Remaining risks

Critic Agent

Role: Evaluates final plan against requirements

Input:

- Replanning agent output
- Assignment instructions
- Rubric

Output:

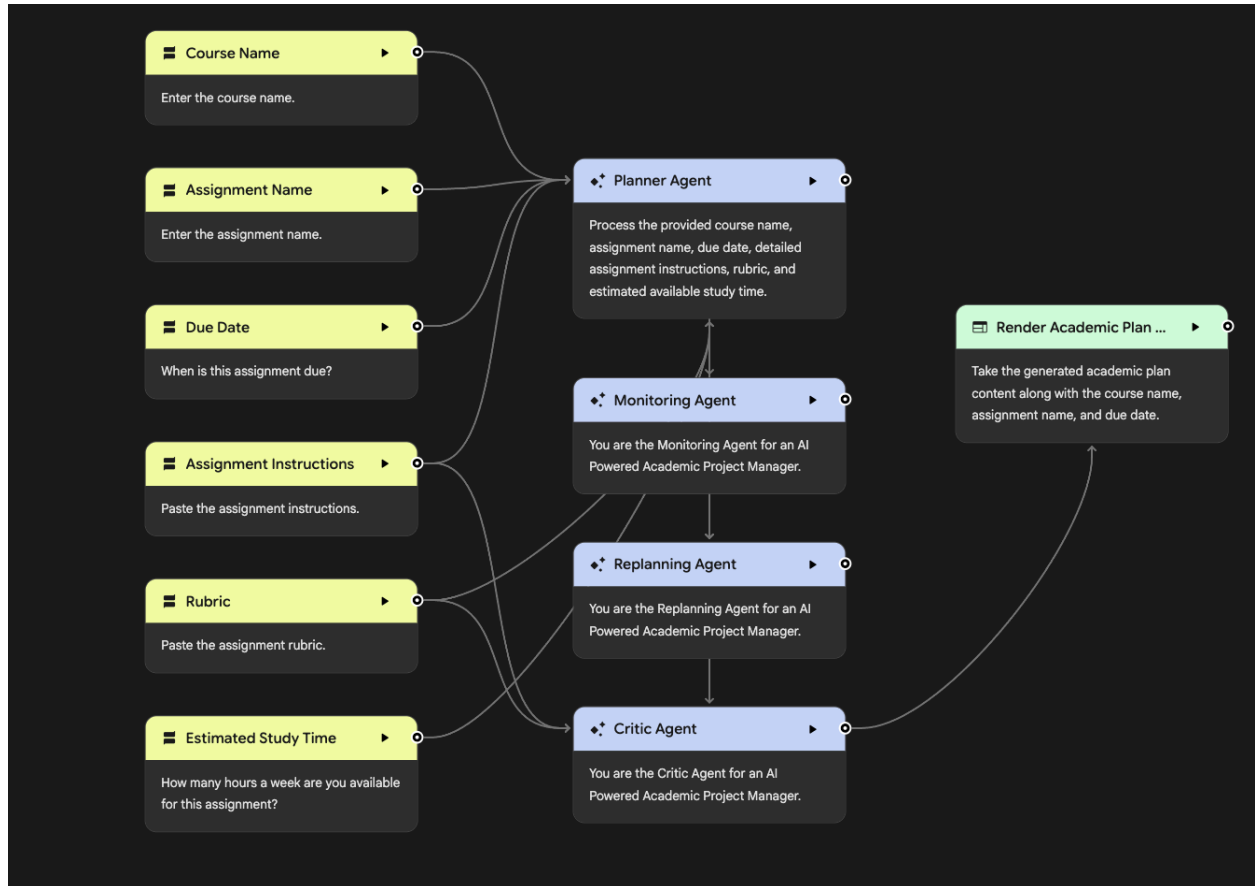
- Strengths
- Weaknesses
- Rubric alignment
- Final recommendations
- Final approved plan summary

Agents communicate through a fixed pipeline. Each agent received the previous agent's output, processed it, and then passed it forward. The monitoring agent identifies issues in the initial plan, the replanning agent corrects those issues, and the critic agent evaluates final quality and alignment.

Test Case

The system was tested using an individual Taxation Analysis project. The assignment required calculating a taxpayer's gross income, deductions, taxable income, tax liability, and providing tax planning recommendations based on a given financial scenario.

Architecture diagram



Multi-agent workflow architecture

Final output ↓

Individual Taxation Analysis

John Smith: Tax Year 2025



FINAL SUBMISSION DEADLINE
March 31, 2025



Assignment Overview

| Strategy Summary

"The plan correctly identifies the necessary flow for tax analysis, starting with research and moving through Gross Income, AGI, and Deductions to taxable income and liability."

TIME EFFORT

5 Hours

DELIVERABLE

2-3 Page Report

TAX YEAR

2025 Federal



Execution Steps

1

Foundation & Research

Calculate Total Gross Income (\$83,150) and AGI by adjusting for Student Loan Interest and IRA Contributions.

2

Deduction Comparison

Compare Itemized Deductions (Charity, Medical, SALT) against the 2025 Standard Deduction. Prepare justifications.

3

Calculations & Credits

Apply 2025 Federal Marginal Brackets to determine liability and check for Credit eligibility (EIC/CTC).

4

Strategy Formulation

Draft 3 specific tax-reduction strategies for future tax years with clear impact explanations.



Priority Watchlist

High Focus

Calculation Accuracy Risk

The 7.5% AGI floor for medical expenses is a critical pitfall. Use a separate spreadsheet for the math to ensure zero calculation errors before transferring results to the report.

SALT Cap Monitoring

Ensure the \$3,500 State and Local Taxes paid is noted as being comfortably under the \$10,000 SALT cap.

Strategic Insight

Final recommendations and rubric alignment analysis.

TOP STRENGTHS

- ✓ Logical Sequential Structure
- ✓ Specific 2025 Data Focus

AREAS TO FORTIFY

- ⓘ Explicitly justify Standard vs Itemized
- ⓘ Check Child Tax Credit eligibility

Rubric Mapping

CALCULATION ACCURACY

60/60 Pts Target

Tax Liability (20 pts)

Tax Credits (10 pts)

Recommendations (10 pts)

Presentation (5 pts)

Final Success Tip

"Since \$3,500 (SALT) + \$1,200 (Charity) + medical expenses is unlikely to exceed the standard deduction, ensure the 'comparison' in Task 2 is very clear and explains precisely **why** the standard deduction is used. This earns you those final presentation points."

Save Plan to PDF



Phase-Based Timeline



MARCH 25 - 26

Phase 1: Foundation

Focus on AGI and Deduction comparison logic. Establish the base numbers.



MARCH 27 - 28

Phase 2: Calculation & Strategy

Apply tax brackets and finalize the narrative for the 3 tax-saving recommendations.



MARCH 29 - 31

Phase 3: Finalization

March 29 is your buffer day. March 30 for compilation. March 31 for final submission.



Target Milestones



MILESTONE 1

Verified 2025 Rates



MILESTONE 2

AGI Calculation Solidified



MILESTONE 3

Strategy Draft Complete



MILESTONE 4

Tables Merged into Word



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